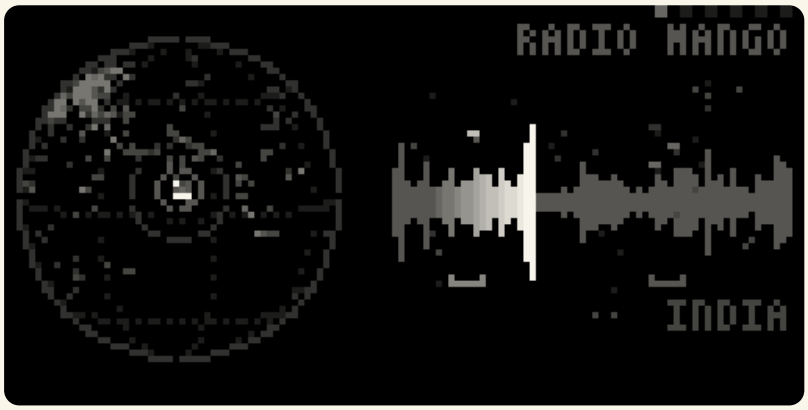
# POLLEN

*the world's radio, carried on the wind, scattered as grains*




player's manual

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pollen tunes into a live radio station somewhere on Earth, catches it on a reel, a loop of the last few seconds like a strip of tape, and scatters it as grains: tiny pieces of the broadcast, played back at their own pitch, length and place. Then the wind carries you on to another station, near or far, and the grains drift from one place to the next.

Pollen grains and radio waves both cross the planet through the sky, invisibly, and land far from where they started. You don't pick stations from a list. You choose how the wind blows, and you shape what it brings.

It's built for listening as much as playing. Leave the wind still and stay with one station, or give it a pace and let it wander the globe on its own while the reel turns every voice, song and jingle into texture.

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# *Foreword*

Pollen was born from the idea of capturing the aleatory nature of radio stations and all the variety that exists in the thousands of radio stations scattered across the globe and treating them as textures for generative music.

The name "pollen" reflects the way the system spreads audio material across grain-like units, scattering it to create evolving textures. It's very much in the regular spirit of glassgarden. creations where nature's patterns and processes inspire the design and behavior of the system.

The initial idea was to capture radios from radio garden, a web platform that lets you explore live radio stations around the world but their audio streams are not accessible via API.

Pollen mixes a source of almost randomic audio with granular synthesis treatment and some effecta in the chain creating a rich and evolving sonic texture.

Hope you enjoy exploring the sonic textures that pollen creates.



*Happy patching!*
*Carlos E.*
*Founder and creator of glassgarden. / Voltage Foundry Modular*

# WHAT YOU NEED

- A **norns** (any kind: the original, shield, or fates), **on the internet**: the stations are live.
- **mpv** and **socat** on the norns, the programs that play the stations. pollen offers to install them the first time you open it: one press of K3.
- Optional: the **mi-UGens**, the Mutable Instruments sound modules for norns. With them, the reverb is MiVerb, the reverb from Clouds and Rings; without them, a plainer one stands in.

To install pollen, type this in the maiden REPL:

```
;install https://github.com/glassgarden/pollen
```

You can also play with pollen on the web at https://pollen.glassgarden.cc

## INSTALLING MPV AND SOCAT

The first time you open pollen, it checks for them. If they're missing, a screen lists them with their sizes (mpv is the big one: it brings its sound libraries along):

- **K3** installs both. It takes a few minutes, with a progress bar; the norns needs to be on the internet. When it's done, pollen starts again, with the radio.
- **K2** is "not now": pollen plays without the radio, scattering whatever comes into the norns input instead, and the screen says what's missing. It asks again the next time you open it, or straight away from **PARAMS > radio > install mpv and socat**.
- **E2** moves through the list to say what each one is for.

If the screen can't install them itself (it needs the norns' usual passwordless `sudo`), it shows the command to type instead. From a computer on the same network:

```
ssh we@norns.local
sudo apt install -y --no-install-recommends mpv socat
```

(the password is the one you set for the norns, `sleep` if you never changed it).

## REMOVING THEM

pollen keeps a note of what it installed, so it can take exactly that away again: **PARAMS > radio > remove mpv and socat** shows what it will remove, and K3 removes it. Do this before you delete pollen if you don't want them anymore: deleting a script doesn't run any of its code. If mpv or socat was already on your norns before pollen, it's left alone, since another script may use it.

Removing mpv leaves the libraries it brought along. To take those away too, over ssh:

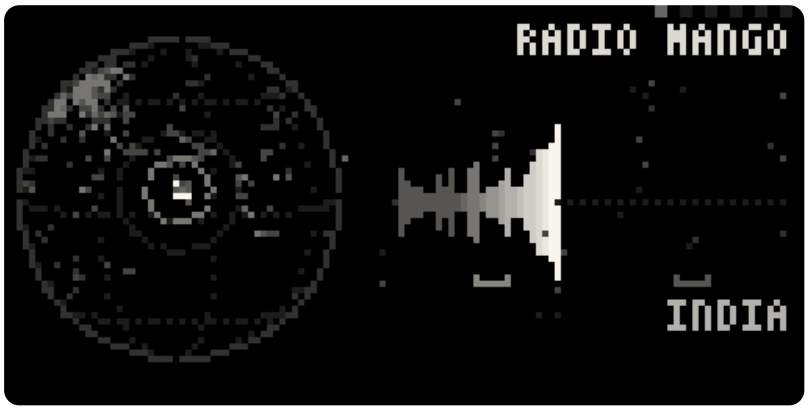
```
sudo apt autoremove
```

That removes every package nothing needs anymore, so read its list before you say yes.

# THE FIRST MINUTES

Open pollen from the norns menu (**SELECT > pollen**). The first time, it fetches the list of stations, the 1,500 most listened-to ones that say where in the world they are, which takes a few seconds. After that the list is kept on the norns and refreshed once a week.

Then pollen tunes into its first station. On the left of the screen, the globe turns to it; on the right, the reel starts filling, drawn as a strip of tape. As you arrive, the station's name and where it is show for a moment.




What you hear is a mix of the station as it plays and the grains scattered from the reel. When you want to move on, choose **wind** with E1 and press **K3**: the wind carries you to the next station.

The wind starts still, so pollen stays with a station until you drift. Give it a **pace** and it moves on by itself.

# PLAYING IT

pollen has **six parts**: wind, catch, scatter, tint, tone and air. You pick a part, then shape it.

| CONTROL | DOES |
|---|---|
| E1 | choose a part: wind, catch, scatter, tint, tone, air |
| E2, E3 | that part's two knobs (its macros) |
| K3 | that part's gesture (drift, press, burst, flip, fade, haze) |
| hold K2 + E2 | that part's variety (which wind, how long a reel, which envelope…) |
| hold K2 + E3 | the heads: how many (catch), how far apart in pitch (tint) |
| hold K1 | peek at every part and the station |
| hold K1 + K2 | back to the station before |

K1 works only when held: a quick tap of K1 opens the norns menu, as in every script.

## CHOOSING A PART

Turn **E1**. The part's name shows at the top left for a moment. Along the top right, a small square for each part, in E1's order, shows which one you're on: its square is the bright one. Left alone for a minute, the squares fade out (the screen is an OLED, and nothing should stay lit for hours); touch anything and they're back.

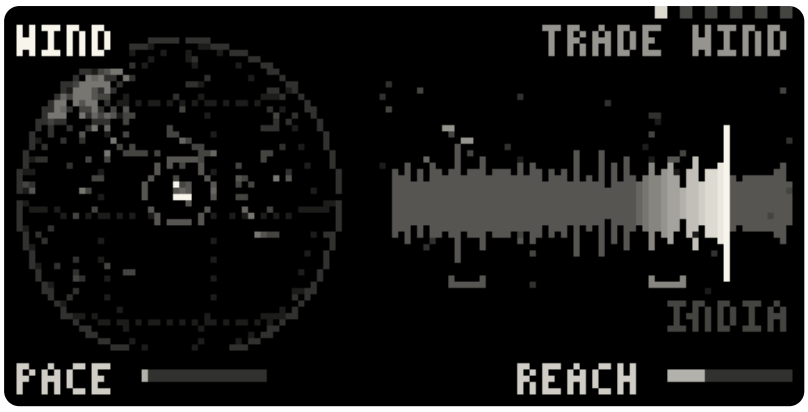
## SHAPING IT

**E2** and **E3** are the part's two knobs. Their names and a bar for where each one is show at the bottom while you turn them: E2's on the left, E3's on the right, where the knobs are on your norns.




## VARIETIES (HOLD K2 + E2)

Each part has a variety: a choice of kind rather than an amount. Hold **K2** and turn **E2**, and its name shows at the top right.




## GESTURES (K3)

**K3** does something right away. Some gestures are moments (a burst of grains, a fade, a drift); others hold until you press K3 again (pressing the reel, flipping the grains, holding the haze). A word on the screen says which.

## PEEKING (HOLD K1)

Hold **K1** to see every part at once, with a bar for each of its two knobs, and the station you're on at the bottom. Let go to come back.




## GOING BACK (HOLD K1 + K2)

Hold **K1** and press **K2** to return to the station before. Do it again to go further back: pollen remembers the last 32 places the wind took you.

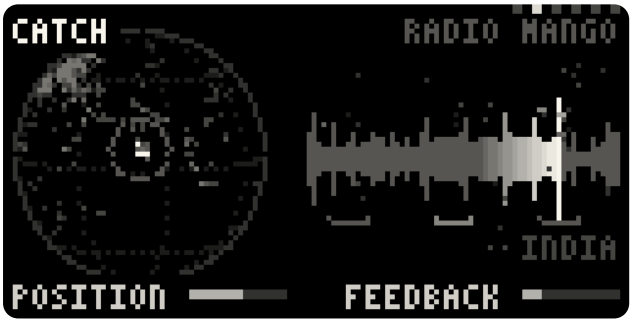
# THE SIX PARTS




## wind

When and where the radio moves to the next station.

- **E2 pace**: how often it moves and selects a new station on by itself. All the way left (where it starts) it moves only when you drift; then every 10, 5, 2 or 1 minutes, or every 30 seconds.
- **E3 reach**: how far a move goes. Left stays close, a city or two away (about 50 km); right crosses oceans and continents (up to 10,000 km).
- **K3 drift**: move on to the next station now.
- **Variety, the wind**: *breeze*, *trade wind*, *eddy*, *gust* or *far side* (see The wind, in more detail).




## catch

The reel: the loop of tape the radio is recorded onto, and the grains are read from.

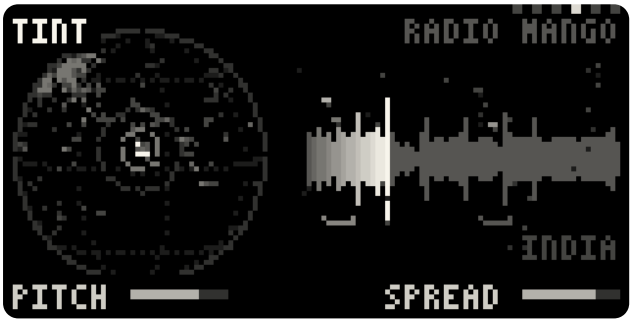
- **E2 position**: where on the reel the grains are read. Left is just behind the recording, so you hear the station almost live; right reaches back into what was caught seconds ago, and the grains come from a wider stretch of it.
- **E3 feedback**: how much of the grains is recorded back onto the reel along with the radio. Left, the reel holds only the radio; right, the grains pile up on themselves into smears and drones.
- **K3 press**: stop recording, so the reel holds what it has and the grains keep playing it. Press again to release it. The station plays on underneath; the reel just stops listening.
- **Variety, reel length**: 4, 8, 15, 30 or 60 seconds of tape (it starts at 15). A new length starts an empty reel.
- **Hold K2 + E3, heads**: how many places on the reel the grains are read from, 1 to 4. The heads sit evenly spread along the reel behind the first (which position moves), so you hear the station now and seconds ago at once. Each has its bracket under the tape.




### scatter

The grains themselves.

- **E2 density**: how many: from one every two seconds to 40 a second. The more there are, the looser they fall in time and the wider they spread in stereo.
- **E3 size**: how long each one is: from 20 ms, a click or a buzz, to 2 s, a long overlapping wash.
- **K3 burst**: four times as many grains, for a moment.
- **Variety, envelope**: the shape of each grain. *Bell* is smooth, *pluck* starts sharp and dies away, *block* is square and clicky.




### tint

The pitch of the grains.

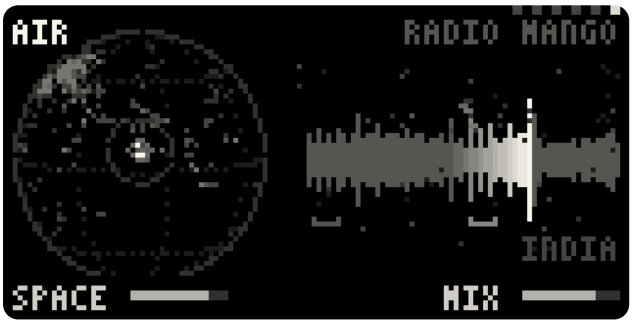
- **E2 pitch**: from two octaves down (left) to two up (right). The middle plays them as recorded. It snaps to the notes of the scale.
- **E3 spread**: how far each grain's pitch wanders from that, up to an octave. Past the middle, some grains also play backwards.
- **K3 flip**: most grains play backwards. Press again to turn them around.
- **Variety, scale**: *octaves*, *fifths* (where it starts), *major*, *minor*, *chromatic*, or *free* (no snapping).
- **Hold K2 + E3, heads apart**: with more than one head, each plays the next note of the scale up from the one before it: in *unison*, or *one to four steps* of the scale apart. In major or minor that's a chord. In free, the steps are semitones. Nothing goes more than two octaves above the first head.




### tone

The color of the sound.

- **E2 color**: one filter knob, like a DJ mixer's. The middle is open. Turn left and a low-pass closes, darker and muffled; turn right and a high-pass opens, thinner. It rings more toward either end.
- **E3 drive**: saturation, from clean to crunchy.
- **K3 fade**: like a shortwave signal fading: for a moment the sound sinks and darkens, then comes back.
- **Variety, media**: what the sound passes through, after Mutable Instruments Beads: *digital* (clean), *vintage* (a lo-fi sampler), *clean tape*, *dirty tape*.

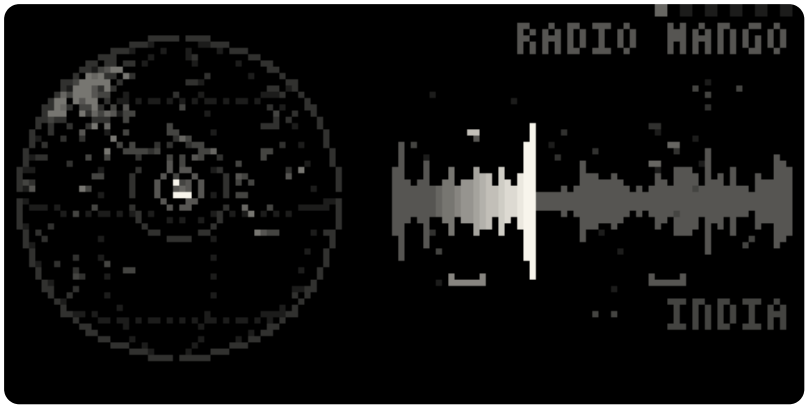



### air

The space around it, and the final mix.

- **E2 space**: how much of the grains goes to the reverb, and how big the room is.
- **E3 mix**: left is the radio alone, as it plays; right is the grains alone; between them, both. It starts mostly grains, with some of the station underneath.
- **K3 haze**: hold the reverb's tail so it hangs in the air. Press again to let it go.

pollen shows pictures, not numbers. The screen has two halves.




**On the left, the globe.** Every station is a dot, so the dots draw the places people broadcast from. The station you're on is at the middle, sending out radio waves as loud as it is. The wind's last few moves are its trail. While tuning, the station being tried blinks at the end of a dotted path and the globe turns toward it.

**On the right, the reel, laid flat like tape.** The waveform is what it caught, with the write head sweeping across it: the newest sound is brightest just behind the head, and what isn't caught yet is a dotted line. The bracket under the tape is where the grains are read from (the catch's position); with more heads, there's a bracket for each. The grains leave the tape from there, upward when they play forwards and downward when backwards, faster the higher they're pitched. Silence throws none. Pressed, the head stops and the whole tape holds evenly lit. The haze over it is the reverb.

The station's name and place show as you arrive, and on a peek.

# THE WIND, IN MORE DETAIL

The wind decides where the next station is. Its variety picks the kind of wind, and **reach** sets how far each move goes.

| WIND | MOVES |
|---|---|
| breeze | hops off in any direction, as far as reach says (where it starts) |
| trade wind | carries on steadily along one bearing, around the world if you let it |
| eddy | a random walk: the bearing turns a little with each move |
| gust | anywhere at all: a country at random, then a station in it |
| far side | to the other side of the world, and back |

A move always lands somewhere: if it would end out at sea, it takes the nearest station far enough along, not one back on the coast it left. Stations already visited are skipped until every one has been heard.

**How a new station comes in.** While the next station loads, the last one keeps playing. Once it's really playing, the two crossfade over one pass of the reel, so the head writes the new station over the old a little more each moment and the grains travel from one place to the other. A station that doesn't play in time is skipped, and the wind tries another.

# THE RADIO'S SETTINGS



In the norns menu (a quick tap of K1), **PARAMS** has every knob, variety and gesture, grouped by part, so they save with presets and can be mapped to MIDI. Two more groups are about the radio and the levels:

**radio**

- **handover**: how a new station comes in. *over the reel* (where it starts) blends it in over one pass of the reel; *pressed* presses the reel while the station loads, then crossfades quickly.
- **static while tuning**: some shortwave hiss while the wind is moving you (off to start).
- **crossfade (pressed)**: how long the crossfade takes with the pressed handover.
- **give a station**: how long a station has to start playing before it's skipped (10 s to start).
- **refresh the stations**: fetch the list again now, instead of waiting for the week.
- **install mpv and socat**: the install screen again, if you said "not now" (see Installing mpv and socat).
- **remove mpv and socat**: take away what pollen installed (see Removing them).

**levels**

- **level**: the output.
- **input**: how loud everything coming in is: the radio, and anything plugged into the norns input, which is scattered along with it (or instead of it, without the radio).
- **static**: hiss and crackle all the time, as if the signal were weak.
- **clear the reel**: empty the tape.

# SAVING WHAT YOU LIKE



pollen picks up where you left off: every setting is kept on the way out and read at the next start. To keep more than one, use presets: in **PARAMS**, **PSET** is at the top (*save*, *load*, *delete*). A preset remembers every knob, variety and gesture, and the radio settings. It doesn't remember the station: the wind takes you somewhere new.

# A FEW PLACES TO START



**Slow tourism.** wind: *breeze*, pace at 1 minute, reach low. catch: position a little right. scatter: density in the middle, size large. air: space high, mix mostly grains. Neighbouring towns' stations drift into each other.

**Around the world.** wind: *trade wind*, pace at 2 minutes, reach all the way right. tint: *fifths*, spread low. Every couple of minutes you're a few thousand kilometres further along.

**Shortwave.** tone: color a little left, drive up, media *dirty tape*; press K3 on tone now and then. PARAMS: static a little up. wind: *gust*.

**A chord from a voice.** catch: heads 3. tint: *major*, heads apart *two steps apart*, spread low. scatter: size large, envelope *bell*. Find a station with someone talking.

**Freeze a moment.** When something good comes by, press K3 on catch: the reel holds it while the station plays on. Turn up feedback for drones, then release when you're ready for the radio again.

# IF SOMETHING SEEMS WRONG



**The screen says pollen needs mpv and socat.** They aren't installed yet: open **PARAMS > radio > install mpv and socat**, or see Installing mpv and socat. Until then pollen plays the norns input.

**The install stopped with an error.** The screen shows the end of what went wrong: K3 tries again, K2 closes. The usual cause is the norns being offline, or apt's package lists being out of date; over ssh, `sudo apt update` and then try again.

**No stations.** The norns couldn't fetch the list: check its Wi-Fi in the norns menu (**SYSTEM > WIFI**), then **PARAMS > refresh the stations**. Without a list, pollen plays the input.

**A station is silent, or the wind keeps moving on.** Some stations are offline, or slow to start. pollen gives each one a few seconds (**give a station** in PARAMS) and then tries another. A slow network makes this happen more: give stations longer.

**Nothing changes when I turn a knob.** Check the mix (air, E3): all the way left is the radio alone, with no grains.

**It's too loud, or clipping.** Lower **level** in PARAMS, or feedback in catch: at its highest the grains pile up on themselves (a limiter keeps them from going over, but they get dense).

# APPENDIX: HOW IT WORKS



## THE SIGNAL PATH

The station plays in **mpv** on the norns, which sends its sound into SuperCollider. There it's written onto the reel, a loop of 4 to 60 seconds, and up to four heads read grains back from behind the write head. The grains go through the filter, the drive and the media, then to the reverb, and are mixed with the station as it plays. A limiter at the end keeps it all under full scale. Two mpv players take turns, so one can load the next station while the other plays.

## WHERE THE STATIONS COME FROM

The list comes from radio-browser.info, a community database of radio stations; pollen keeps the 1,500 most listened-to ones that say where they are. The streams are the stations' own, played live and never stored.

## IN A BROWSER

pollen also runs in a web browser at https://pollen.glassgarden.cc/: the same script, with the engine rebuilt for the web. There, only stations whose servers allow a web page to play them can be heard (most do). Behind the device, the page shows a field of grain under the sky at the hour it is where the station plays, with pollen drifting on the wind.

# CREDITS



pollen is made by glassgarden. It's free software under the GNU GPL v3.0.

- Ideas from Émilie Gillet's Mutable Instruments Clouds and Beads (MIT): the media stage follows Beads' four. Inspired by Make Noise Morphagene and Endorphin.es Evil Pet.
- The reverb, when installed, is MiVerb from Volker Böhm's mi-UGens.
- The stations come from radio-browser.info.
- mpv and socat are installed and removed with vehka's norns-deps (MIT).
- The browser version runs on norns-web and wasmoon.

Mutable Instruments, Make Noise, Endorphin.es and monome are trademarks of their owners, named only to say what pollen was inspired by or works with.